

***MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY
RESOURCE MANAGEMENT-WATER RESOURCES DIVISION
POINT SOURCE MONITORING***

Report of a
Municipal Wastewater Survey
Conducted at
Cadillac WWTP
NPDES No. MI0020257
Wexford County
Cadillac, MI
October 14, 2014

SURVEY SUMMARY

Wastewater monitoring was performed during one twenty-four hour survey period starting Tuesday, October 14, 2014.

The results of this survey are compared to the final limitations in the facility's National Pollutant Discharge Elimination System (NPDES) Permit, No. MI0020257, in Table 4.

The results of this survey are compared to the results reported by the facility in their October Discharge Monitoring Report (DMR) in Table 4.

A composite sample from monitoring point (001A) was split with the facility to compare laboratory analyses. The results are shown in Table 5 as 001A-F.

The results of this survey are compared to the results of the previous survey performed in January, 2011 in Table 6.

The results of the physical, chemical and bacteriological analyses are presented in Tables 1 & 2.

There is no Table 3 in this report.

SURVEY PROCEDURE

The flow and samples were obtained as follows:

Sample Location	Flow Measurement	Sampling Methods
001A samples collected from plant's automatic composite sampler post UV disinfection, excepting fecal coliform, available cyanide, pH, Temp., and D.O., which were sampled from pipe, post UV just prior to discharge.	Totalizer reading.	Grab composite and individual grab samples.
001A-F samples collected from plant's automatic composite sampler, post UV disinfection.	-----	Split plant's automatic composite sample.
Upstream samples collected from the Clam River @ Smith St. bridge. (N44.26330,W085.40099)	-----	Individual grab samples.

A grab composite consists of a series of individual grabs composited into one sample.

An individual grab is a single instantaneous sample.

Samples were analyzed by the MDEQ Laboratory located in Lansing and TriMatrix Labs of Grand Rapids, MI.

Samples were preserved according to Table 7. The results of the physical, chemical, and bacteriological analyses are presented in Tables 1 & 2. Letter codes for laboratory results are defined in Table 7.

There is no Table 3 in this report.

TABLE 1 - Analyses of Composite Samples

Monitoring Point

Survey Period From:

To:

Computed Flow Rate (MGD):

001A

10/14/2014 10:10

10/15/2014 10:10

1.5

	<u>mg/L</u>	<u>lbs/day</u>
Suspended Solids	7	88
CBOD-5	4.8	60
NH3	0.31	3.9
Total Phosphorus	0.15	1.9
Alkalinity (as CaCO3)	130	NA
Hardness (as CaCO3)	250	NA

	<u>ug/L</u>	<u>lbs/day</u>
Total Copper (Cu)	7.3	0.091

TABLE 2 - Analyses of Grab Samples

Monitoring Point	001A	001A	001A	Upstream
Matrix	Water	Water	Water	Water
Date Collected	10/14/2014	10/14/2014	10/15/2014	10/14/2014
Time Collected	10:15	15:10	10:25	11:00
	<u>mg/L</u>	<u>mg/L</u>	<u>mg/L</u>	<u>mg/L</u>
Temperature (F)	64	62	63	55
pH (S.U.)	7.0	7.0	7.0	6.9
Dissolved Oxygen	8.1	8.1	7.9	---
Available Cyanide	<0.002 ND,U	---	---	---
Alkalinity (as CaCO3)	---	---	---	70
Hardness (as CaCO3)	---	---	---	75
Fecal Coliform (cts/100 ml)	<10 H	<10 H	10	---

TABLE 4
Comparison of the Survey Results with the Facility's NPDES Permit and Discharge Monitoring Report

Parameter (unit)	Final Effluent Limitations				October Discharge Monitoring Report						Survey Results	
	Maximum Monthly	Maximum 7-day	Minimum Daily	Maximum Daily	Average	Maximum 7-day	Minimum Daily	Maximum Daily	10/14/2014	10/15/2014	Composite	(grab)
<u>Monitoring Point (001A)</u>												
Flow (MGD)	(report)			(report)	1.91			2.12	2.12	1.97	1.5	
CBOD-5 (mg/L)	7			10	8.34			21.45	4.2	6.2	4.8	
CBOD-5 (lbs/day)	190	270			133.25	122.14			74.08	101.11	60	
Suspended Solids (mg/L)	20	30			19.0	16.3			2.0	1.60	7	
Suspended Solids (lbs/day)	530	800			301.5	263.2			35.28	26.31	88	
Ammonia (mg/L)	0.9			2.0	0.7			2.3	1.52	1.88	0.31	
Ammonia (lbs/day)	24	53			11.7	10.7			26.81	30.91	3.9	
Total Phosphorus (mg/L)	0.5				0.73				0.14	0.15	0.15	
Total Phosphorus (lbs/day)	13				11.94				2.47	2.47	1.9	
Available Cyanide (mg/L)	0.0059			(report)	0.002							(<0.002 ND,U)
Available Cyanide (lbs/day)	0.16				0.0332							
Fecal Coliform (cts/100 ml)	200	400			168.4	654.0			5	8		(<10 H, <10 H, 10)
pH (S.U.)			6.5	9.0			7.0	7.6	7.6	7.3		(7.0, 7.0, 7.0)
Dissolved Oxygen (mg/L)			6.0				6.9		7.7	7.8		(8.1 , 8.1 , 7.9)
Total Copper (ug/L)				(report)				50		< 50	7.3	

Table 5
Comparison of the Laboratory Analytical Results
Obtained by the Facility and the MDEQ Authorized Labs

Monitoring Point

Sample Type

Date Collected

Time Collected

001A-F

Composite

10/15/2014

7:00

	-----	-----
	Facility	MDEQ
	<u>mg/L</u>	<u>mg/L</u>
Suspended Solids	1.6	6
CBOD-5	6.15	5.5
NH3	1.88	1.4
Total Phosphorus	0.160	0.15

Table 6 - Comparison of Results with Previous Survey

Monitoring Point		001A
Matrix		Water
Survey Period From:	01/24/11	10/14/14
To:	01/25/11	10/15/14
	<u>mg/L</u>	<u>mg/L</u>
Suspended Solids	4	7
CBOD-5	1.7 J	4.8
NH3	<0.1 ND,D	0.31
Total Phosphorus	0.13	0.15
Alkalinity (as CaCO3)	110	130
Hardness (as CaCO3)	257	250

TABLE 7 - Sample Preservation

Parameter	Preservative
COD/TOC/Phenol/Nutrients (Chlorine Absent)	5 drops conc. H ₂ SO ₄ /250 ml (to pH <2)
Available Cyanide	Treated with Lead Carbonate, filtered and preserved with Sodium Hydroxide to a minimum pH of 12.
Dissolved Oxygen (D.O.)	Meter reading.
Total Metals	2 ml 1:1 HNO ₃ /250 ml (to pH <2).
Microbiology	2 drops 10% sodium thiosulfate/125 ml to dechlorinate sample.
Samples preserved as required, cooled to 4 degrees Centigrade with chain of custody maintained.	

Lab Letter Codes

D - Analyte value quantified from a dilution(s); reporting limit (RL) raised.

H - Sample had exceeded EPA holding time for this analysis when received in the laboratory.

J - Result was detected between MDL and RL and is considered estimated. (TriMatrix Labs)

U - Result was not detected above the MDL. (TriMatrix Labs)

ND - Indicates compound analyzed for but not detected.

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Laboratory Analyses by: Michigan Department of Environmental Quality Laboratory
TriMatrix Labs of Grand Rapids, MI

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